CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 462 Practice

Midterm 462 · 25 marks · a graded student paper

Muharib

01 About this midterm

The midterm of term 462: five questions of five marks each. The questions are copied from photos of a student's paper graded by the instructor, so the answers here are taken from that grading.

Question	1	2	3	4	5	Total
Marks	5	5	5	5	5	**25**

Why this paper matters

The instructor's grading on it settles three points that were open in the 472 and 451 midterms: the altered MERGESORT equation is false, the activity-selection item is "Other value", and "B-tree of degree 3" means at most 3 children and at most 2 keys.

02 Questions

Q1 LINEAR PROBING · 5 PTS

Solve the next problem based on the Linear Probing Algorithm:
Insert keys: 76, 93, 40, 47, 10, 55 within an array with size 7 using the hash function: h(key) = key mod N

Key	N	Hash value
76		
93		
40		
47		
10		
55		

0	1	2	3	4	5	6

Q2 TRUE / FALSE + MCQ · 5 PTS

Choose the correct answer by selecting True or False:

1) The code length in Huffman codes depend on the frequency of occurrence of characters: a. True b. False

2) Greedy algorithm can solve all the dynamic programming problems. a. True b. False

3) An algorithm which uses the future results and uses them to find new results is a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

4) The time to run MERGESORT on a list of size n, $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(\frac{MERGE}{2})$ a. True b. False

5) The recursive activity-selection problem algorithm takes a time of O(n log(n)) a. O(n log(n)) b. $O(n^2)$ c. Other value

6) When dynamic programming is applied to a problem, it takes far less time as compared to other methods that don't take advantage of overlapping sub-problems. a. True b. False

7) f(n) is ω(g(n)) if is asymptotically _____ to/than g(n) a. Greater than or equal b. Less than or equal c. Strictly less d. Strictly greater

8) What is the time complexity of the following function?

```
void func(int n) {
   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         cout << i << "," << j << endl;
      }
   }
}
```

A) O(n) B) O(log n) C) $O(n^2)$ D) O(n log n)

9) What is the best-case time complexity of linear search? A) O(1) B) $O(n^2)$ C) O(n) D) O(log n)

10) Which of the following functions has O(log n) complexity?

a) A function with two nested loops iterating n × n times

b) A function that uses binary search

c) A function that iterates n times

d) A function with three nested loops iterating n times each

Q3 GREEDY · KNAPSACK · 5 PTS

A spice store is allowing you to fill a bag that can hold 5kg with your choice of spices.

Saffron: 2 kgs, SAR 4000, Cardamom: 2 kgs, SAR 1000, Cinnamon: 5 kgs, SAR 500, Pepper: 10 kgs, SAR 500.

Using the greedy algorithm for the Knapsack problem, how should items be chosen to maximize the total value without exceeding the bag's weight limit? And what is the total value?

Item ..... weight ..... Total value: .....

Where is the problem with Greedy algorithms and how to solve it from your point of view?

Q4 LCS · 5 PTS

Using the dynamic programming approach, find the longest common subsequences between two strings X = ACBAED and Y = ABCADF.

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

Q5 B-TREE · 5 PTS

Assume you have an empty B-tree of degree 3. Draw the insert operation for the following elements:

F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y, D, Z, E

Note: Subject to the alphabetical order of the letters.

Display the final result of inserting the keys, use the following tree.

